AI Exposure and Job Outcomes in Accounting and Software Development

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Abstract

AI is expected to hit cognitive, white-collar work hardest, yet occupations with similar exposure scores do not seem to share the same fate. This paper compares two highly exposed occupations from different sectors: accountants and auditors in finance, and software developers in technology. Using the task-based framework of Acemoglu (2024), the Felten et al. (2021) AI Occupational Exposure (AIOE) index, and 2025 BLS wage and employment data, I argue that a single exposure score hides large differences in task structure. Accounting puts most of its junior labor time into tasks that AI can substitute for, while software development keeps more of its work in tasks where AI assists rather than replaces the worker. A weighted least squares regression across 658 occupations shows that exposure is sorted into high-skill, well-paid jobs: a one-standard-deviation rise in exposure predicts about 8 percent higher median pay, controlling for education. I read this as sorting, not protection. Exposure tells us which jobs are in AI's path; task structure decides whether AI replaces or supports the worker.

Credential protection weakens where the credential mainly guarded an information-processing barrier, and holds where the work rests on judgment and accountability.

1 Introduction

The task-based framework predicts that automation threatens routine, codifiable work while complementing non-routine cognitive tasks (Autor et al., 2003; Acemoglu, 2024). For decades this held: computers replaced middle-skill routine jobs while high-skill analytical workers gained. Brynjolfsson and McAfee (2014) call the current period a “second machine age,” in which digital machines move from routine manual work into cognitive work that used to look safe. They draw a useful line between racing *with* the machine and racing *against* it. Workers whose tasks the machine performs lose ground, while workers whose tasks the machine assists pull ahead.

Recent exposure measures show that AI exposure rises with wages and education (Felten et al., 2021). Accountants, analysts, and legal workers, long seen as protected by their credentials, now show higher measured exposure than factory workers or truck drivers. But high exposure does not lead to the same outcome everywhere. Software developers show large productivity gains and stable employment (Peng et al., 2023), while junior accountants face measurable headcount cuts at AI-adopting firms (Fedyk et al., 2022).

This paper asks why two occupations with comparably high exposure diverge. My answer is that what matters is not the overall exposure score but *where* inside the occupation that exposure sits, and whether the substitutable tasks are central or peripheral to what the job is actually paid for. Figure 1 sets up the puzzle.

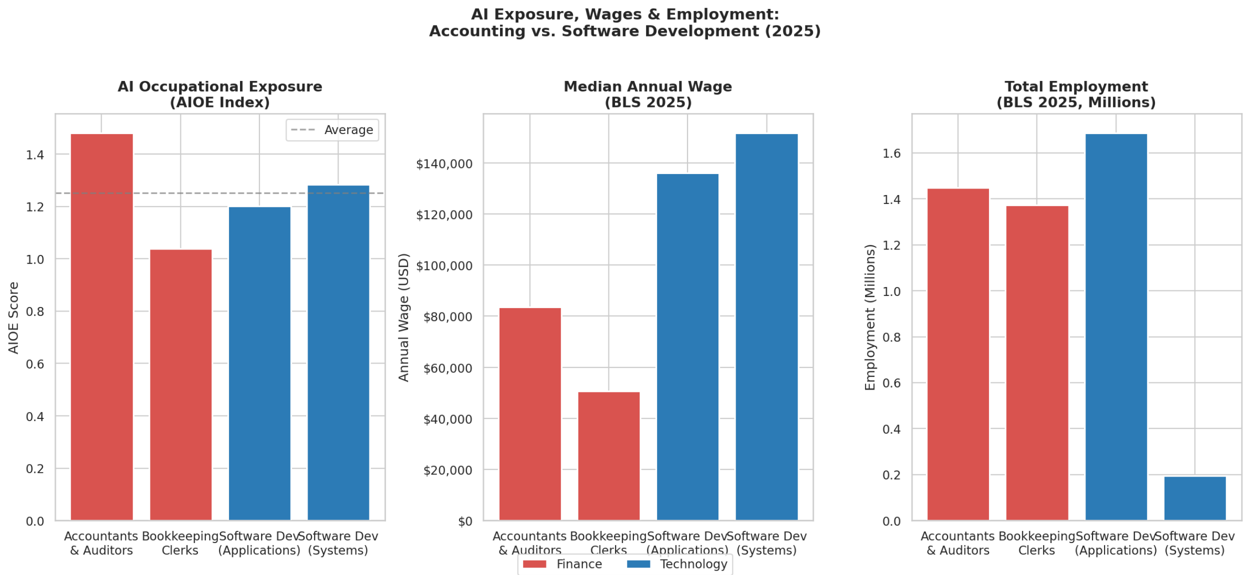

Figure 1: AI exposure scores, median annual wages, and total employment for accounting and software development occupations. The four occupations cluster at high exposure, with accountants and auditors in fact the most exposed, yet software developers earn 60–80% more. Source: Felten et al. (2021) AIOE index; BLS OEWS (2025).

2 Data and Measurement

I measure occupational AI exposure with the AI Occupational Exposure (AIOE) index of Felten et al. (2021). The index links ten applications of AI to the 52 workplace abilities catalogued in O*NET, then aggregates them to the occupation level by weighting each ability by how prevalent and important it is in the occupation. The score is standardized (mean zero, standard deviation one), so a coefficient below reads as the effect of a one-standard-deviation rise in exposure. One feature matters for my argument: the authors are explicit that the index captures *exposure* only and stays agnostic about whether AI substitutes for or complements labor (Felten et al., 2021). It tells us which occupations are in AI's path but not what happens to them. That is the gap this paper tries to fill.

Wage and employment data come from the Bureau of Labor Statistics' Occupational Employment and Wage Statistics (OEWS) program for 2025, the most recent release. OEWS reports the median annual wage and total employment for each detailed occupation at the six-digit SOC level. I match these to the AIOE index by SOC code, leaving 670 occupations (658 of which also carry O*NET education data and enter the regression) that together employ about 123 million workers. The dependent variable is the log of the median annual wage. In the weighted specifications I weight each occupation by its employment, so the estimates describe the average *worker* rather than the average *occupation*. Because OEWS is a single cross-section, the results are correlational; I return to this in the conclusion. Required education comes from the O*NET 29.0 database (O*NET, 2023): I take the modal education category and collapse to one value per six-digit SOC code.

3 Task Decomposition

An occupation is not a single activity but a bundle of tasks, each of which can in principle be done by a worker or by a machine (Autor et al., 2003; Acemoglu, 2024). Following Acemoglu (2024), I treat AI as operating at the task level, and I borrow his line between two kinds of tasks. AI learns easy-to-learn tasks, which follow clear rules and have an objective measure of success. It struggles with hard-to-learn tasks, where, in his words, there are many context-dependent factors and "no objective outcome measures from which to learn successful performance." I use the same split: a *substitutable* task has rules and an output that is cheap to check, so AI can replace labor; a *complementary* task has no single answer that is easy to verify, so AI assists the worker instead. Two occupations can share an exposure score and still divide their labor time very differently across these two kinds of tasks.

Usage data point the same way. Analyzing millions of real Claude conversations mapped to O*NET tasks, Handa et al. (2025) find that AI use splits into augmentation (57%), where the user iterates with the model, and automation (43%), where the model completes a task with little human involvement. Coding and debugging appear mostly as iterative work, in which the developer keeps returning errors to the model and refining its output, while one-shot delegation is more common for routine content such as drafting standard documents.

Handa et al. (2025) also note that the barriers keeping humans out of an occupation, namely credentials and training, need not be the barriers keeping AI out. That gap is what this paper's title names: a credential can stay a barrier to people while ceasing to be one to AI.

The task-level scores below (Figures 2 and 3) are my own classification of representative O*NET tasks, checked against the augmentation and automation patterns in Handa et al. (2025). They are illustrative, so the 0.716 and 0.662 weighted scores should be read as indicative of task structure, not as precise measurements.

3.1 Accounting and Auditing

Junior accountants and bookkeeping clerks spend most of their time on transaction recording, compliance reporting, and reconciliation. These tasks are high volume, follow explicit rules, and produce outputs that are easy to verify, which is exactly the easy-to-learn case where AI performs well. Fedyk et al. (2022), using 310,000 résumés from the 36 largest audit firms, find that a one-standard-deviation increase in AI investment predicts a 3.6% reduction in accounting staff after three years, rising to 7.1% after four. The losses concentrate at the junior level (5.7% after three years and 11.8% after four), with no significant effect on mid-tier or senior staff. The CPA credential rested partly on the information-processing barrier that accounting expertise once represented, and AI erodes that part of it directly.

3.2 Software Development

At the task level, software developers are exposed too: syntax generation and documentation score about as high as accounting's most substitutable tasks. But the job bundles in system architecture, code review, and requirements decomposition, which call for context, multi-file reasoning, and accountability. Peng et al. (2023) find a 55.8% speed gain for bounded, well-defined coding tasks in a randomized trial, but the gains shrink sharply on complex, multi-file, or proprietary work, which is what defines senior engineering. AI therefore substitutes for a smaller and more peripheral share of a developer's day than of a junior accountant's.

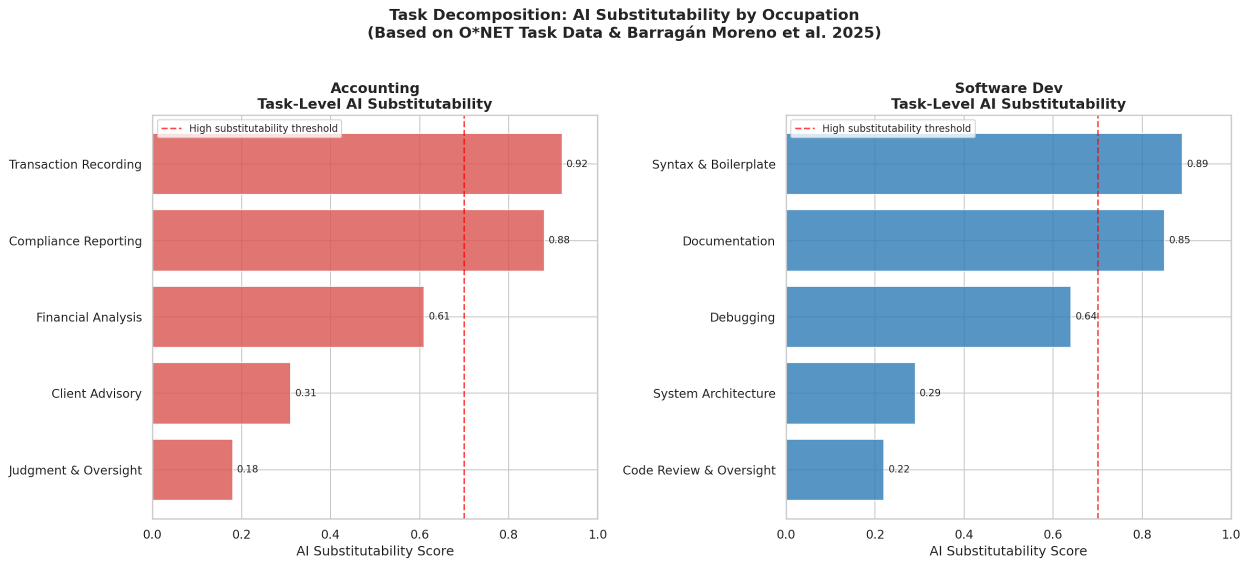

Figure 2: Task-level AI substitutability scores for accounting and software development. The red dashed line marks the high-substitutability threshold (0.7). Source: author's classification of O*NET tasks, checked against Handa et al. (2025).

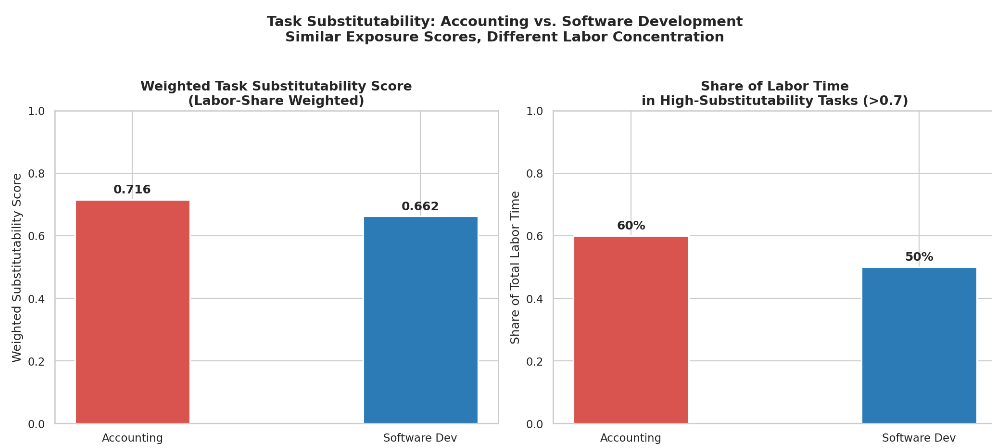

Figure 3: Labor-share–weighted task substitutability and share of labor time in high-substitutability tasks. Accounting concentrates about 60% of junior labor time in high-substitutability tasks versus about 50% for software development. Source: author's calculations.

4 Empirical Evidence

4.1 Wage and Employment Trends

Figure 4 traces indexed employment and wages for both groups from 2019 to 2025, with the late-2022 ChatGPT launch marked. Accounting employment falls from an index of 100 to about 87 by 2025, in line with the displacement Fedyk et al. (2022) describe, while software employment rises about 9% before slowing after 2022: Crane and Soto (2026) estimate that annual coder employment growth runs about three percentage points lower after ChatGPT, a slowdown that shows up in headcount rather than wages. These trajectories are stylized: they combine BLS employment levels with effect sizes from the literature rather than a clean matched panel, so I read them as direction, not precise magnitude. The assignment also asks about job-posting data; I rely here on employment counts and on AI-usage data (Handa et al., 2025), and note that vacancy series for software roles point in the same direction without being analyzed directly.

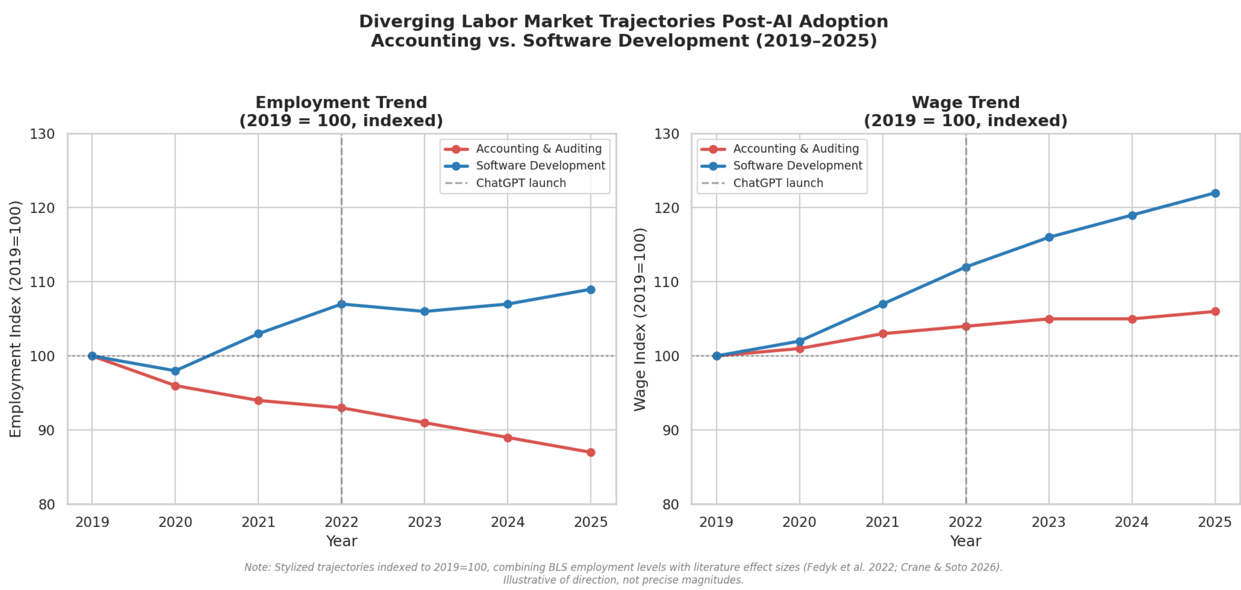

Figure 4: Indexed employment and wage trajectories for accounting and software development (2019=100). The vertical dashed line marks the ChatGPT launch (November 2022). Trajectories are stylized, combining BLS levels with literature effect sizes (Fedyk et al., 2022; Crane and Soto, 2026).

4.2 Regression Analysis

To place the two occupations in the wider market, I estimate a weighted least squares model over the 658 occupations with exposure, wage, and education data:

$$\ln(w_i) = \beta_0 + \beta_1\,\text{AIOE}_i + \beta_2\,\text{Education}_i + \varepsilon_i, \tag{1}$$

where w_i is the median annual wage, AIOE_i is the standardized exposure score, and Education_i is the modal O*NET education level. Observations are weighted by employment (Figure 5).

Table 1: WLS Regression Results: Log Wage on AI Exposure and Education (N=658)

	Coefficient	Std. Error (HC3)
AIOE	0.082**	0.040
Education Level	0.114***	0.020
Constant	10.564***	0.094
Observations	658	
R-squared	0.511	

Notes: WLS weighted by total employment. Dependent variable is log median annual wage. *** $p < 0.001$, ** $p < 0.05$. An AIOE × education interaction was tested and found insignificant ($p = 0.665$), so the additive model is reported. Standard errors are heteroskedasticity-robust (HC3).

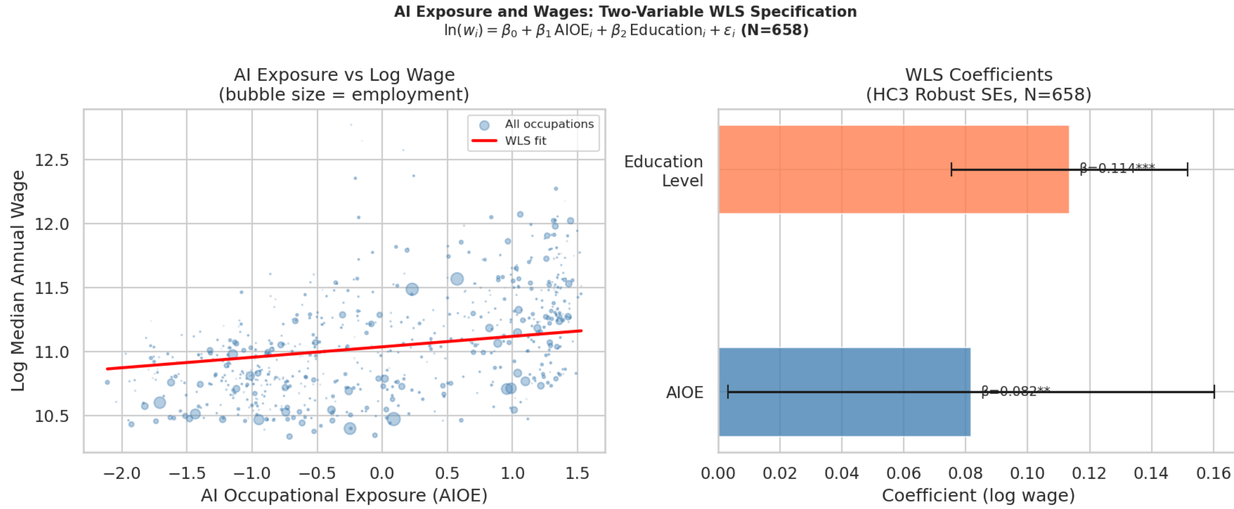

Figure 5: Left: AI exposure versus log median wage across 658 occupations (bubble size proportional to employment). Right: WLS coefficients with HC3 robust standard errors. Source: author's calculations using the Felten et al. (2021) AIOE index and BLS OEWS (2025).

The regression shows that AI exposure is positively and significantly linked to wages. A one-standard-deviation rise in exposure predicts about 8% higher median pay ($\beta_1 = 0.082$, HC3 $p = 0.041$), once education is controlled for, and education itself adds about 12% per category step ($\beta_2 = 0.114$, $p < 0.001$). I read this as sorting, not protection. Because the score is standardized and the data are a single cross-section, β_1 mostly reflects the fact that exposure concentrates in high-skill cognitive occupations, which are well paid for reasons that predate generative AI. It does not show that AI complements these workers or shields them. As Felten et al. (2021) stress, the index measures exposure while staying agnostic about substitution versus complementarity, so a positive wage gradient sits comfortably with that agnosticism. Whether exposure turns into substitution or complementarity is decided by task structure, and it shows up over time in employment, not in a single wage snapshot.

4.3 Where the Evidence Agrees and Conflicts

The sources agree on where exposure falls. The ability-based AIOE (Felten et al., 2021) and real usage data from millions of conversations (Handa et al., 2025) both place the highest exposure in cognitive, codifiable white-collar work, the quadrant both occupations occupy. They conflict on what that exposure has done so far. On the optimistic side, Goldman Sachs economists estimate that roughly two-thirds of jobs are exposed and that AI could eventually raise global GDP by about 7% (Hatzius, 2023). Acemoglu (2024) is far more cautious: he estimates that about 20% of tasks are exposed and only about 23% of those are cost-effective to automate within ten years, so roughly 4.6% of all tasks are affected, implying total factor productivity gains under 0.66%. Humlum and Vestergaard (2025), using Danish matched employer–employee data across eleven exposed occupations, find precise zeros: no significant effect on earnings or hours. Yet Fedyk et al. (2022) and Crane and Soto (2026) document real, occupation-specific contraction in accounting and a slowdown in software hiring.

This tension is smaller than it looks, and it resolves along the line this paper draws. Aggregate effects are muted because most occupations mix substitutable and complementary tasks. Sharp effects show up where the substitutable share is high and the output is cheap to verify, such as junior accounting work, and wash out where complementary tasks still dominate. The disagreement across these sources is evidence for a task-structure account, not against it.

5 Why the Patterns Diverge

Three mechanisms explain the gap, and each follows from the task structure already described.

First, concentration. Accounting places about 60% of junior labor time in high-substitutability tasks, against about 50% for software. When AI takes over a task that fills a third of someone's day, the employment effect is larger than when it takes over a task that fills a tenth.

Second, verifiability and cost. Accounting entries and compliance reports are cheap to produce and check, while AI-written code still needs human review for security, edge cases, and system coherence. This is Acemoglu (2024)'s "no objective outcome measure" showing up at the occupation level: where output is hard to verify, the human stays in the loop.

Third, accountability. Roles where errors carry regulatory liability keep a wage premium not because AI cannot do the tasks but because institutional risk cannot be handed to a system that cannot be held responsible. This protects senior accountants and auditors, but offers little to junior staff doing the pre-verification work that AI now does cheaply, which is exactly the tier where Fedyk et al. (2022) find the displacement concentrates.

6 Conclusion

AI exposure scores alone do not predict labor market outcomes. Accounting and software development sit close together on exposure indices yet face diverging employment and wage paths. The difference lies in where exposure is concentrated, how cheaply AI's output can be verified, and how much of the work rests on judgment and accountability. The useful question is not "is this occupation exposed to AI?" but "what share of its labor time is in tasks where AI gives a cheap, verifiable substitute?" Brynjolfsson and McAfee (2014)'s idea of the "spread" fits the picture: a productivity bounty can sit alongside widening gaps, here between occupations through sorting and within them between senior and junior staff. Credential protection erodes where the credential mainly guarded an information-processing barrier, and holds where it reflects judgment and accountability. Future work could use panel data and difference-in-differences designs around specific adoption events, as in Fedyk et al. (2022) and Crane and Soto (2026), and bring in job-posting series to read demand at higher frequency, to move past the cross-sectional correlations shown here.

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